

Announcements

- Next week is a day lab - we will be meeting in **LSE-S04** at **3:30 p.m.**

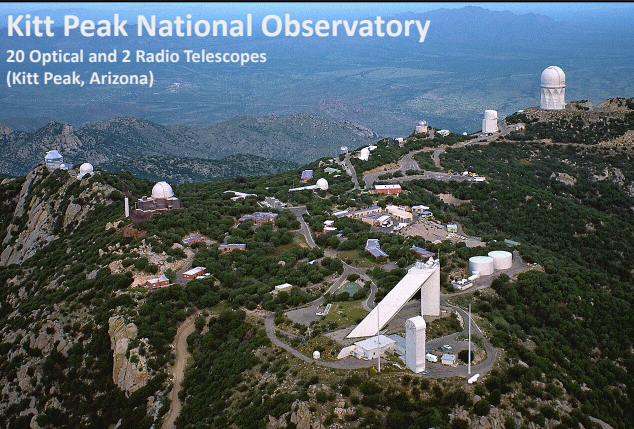
Intro to Telescopes

Telescopes in Arizona

- The best areas to put telescopes are those with
 - High altitudes (i.e. mountains)
 - Dry air (i.e. deserts)
 - And clear skies
- Arizona has plenty of all of those

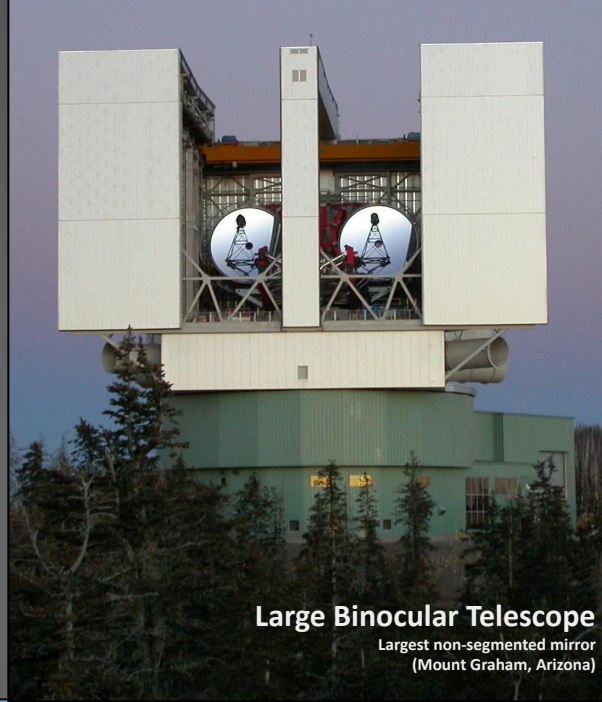
Kitt Peak National Observatory

20 Optical and 2 Radio Telescopes
(Kitt Peak, Arizona)



Lowell Observatory

Where Pluto was discovered
(Flagstaff, Arizona)



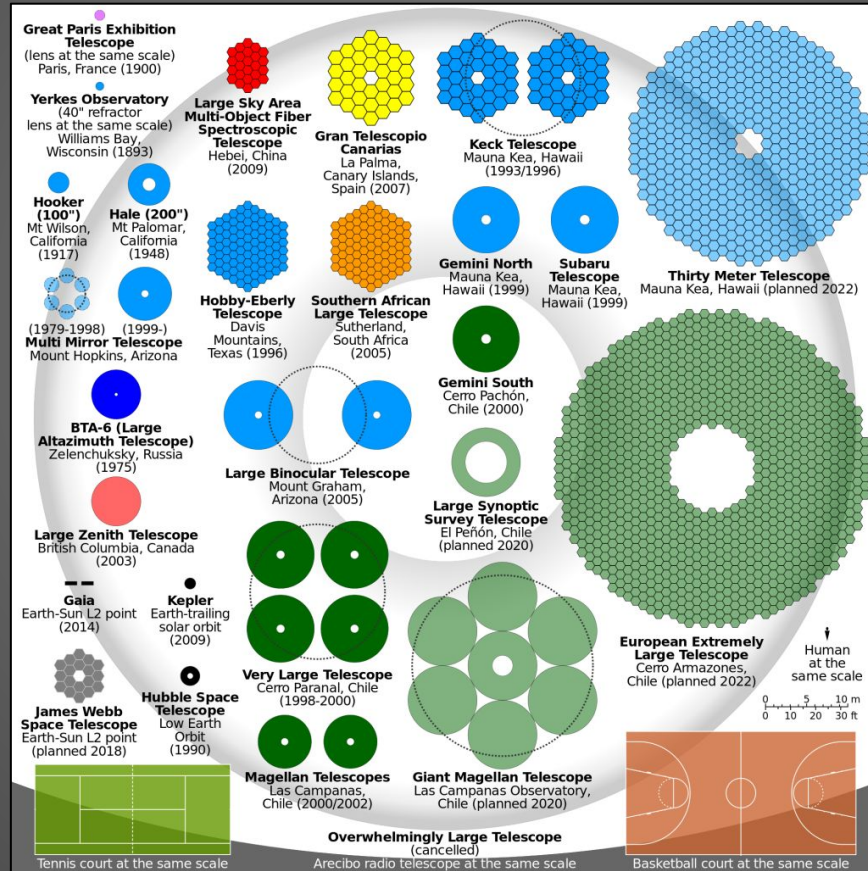
Large Binocular Telescope

Largest non-segmented mirror
(Mount Graham, Arizona)

L
Very Energetic Radiation Imaging Telescope Array System
Four 12m telescopes separated by 100m
(Mount Hopkins, Arizona)

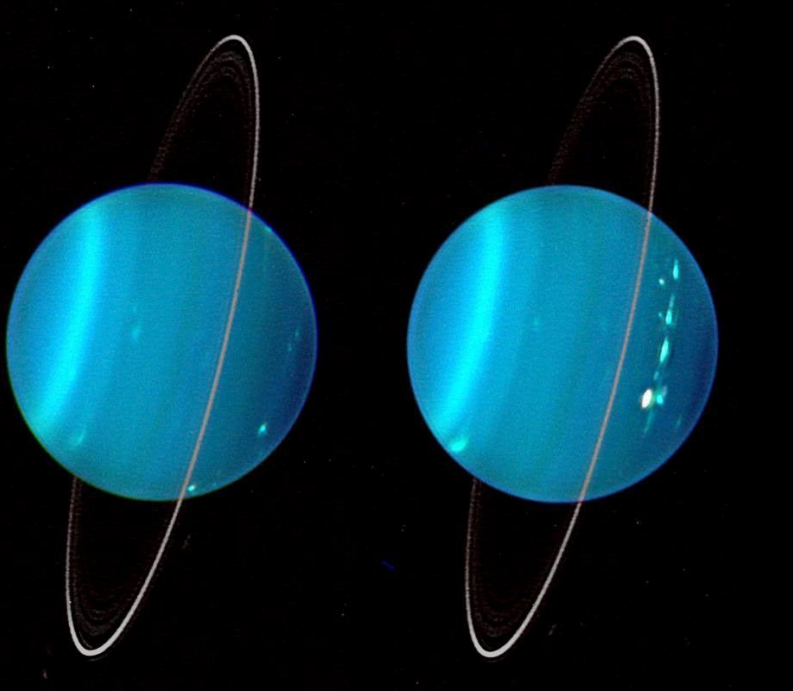


And Around the World



And Around the World

- Hawaii (Mauna Kea)
 - Keck Observatory - Two telescopes, tied for third largest mirror



And Around the World

- Spain (Canary Islands)
 - Gran Telescopio Canarias
 - Current largest single mirror in the world



M 27 - The Dumbbell Nebula
Gran Telescopio CANARIAS (GTC)



And Around the World

- Chile (Atacama Desert)
 - Vera Rubin Observatory
 - Largest Camera in the world
 - Extremely Large Telescope (ELT)
 - Under construction, will be the largest mirror when completed
 - Very Large Telescope (VLT)
 - Largest total mirror area (four mirrors combined)

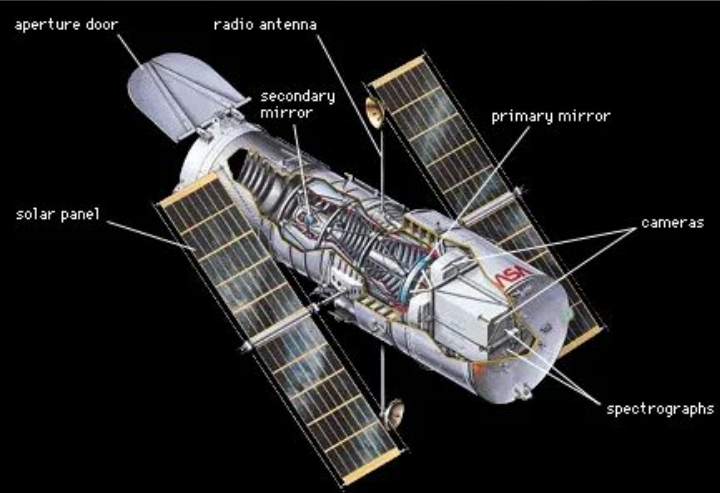


And Around the World



And *Around* the World

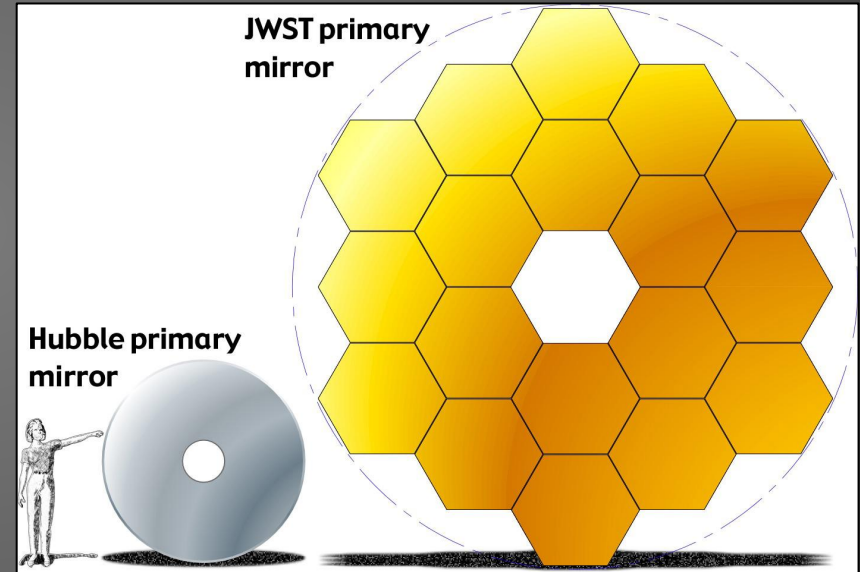
- In Space
 - Hubble (LEO)
 - James Webb (L2)
 - Nancy Grace Roman (L2)
 - Launched in August!
 - And more!



Properties of Telescopes: Light Gathering Power

- Think area
- A good analogy is a bucket collecting rainwater
 - The bigger the area, the more rain (or light) you are going to get

$$A = \pi \left(\frac{D}{2} \right)^2$$



Mirror Diameters:
Hubble: 2.4m
JWST: 6.5m

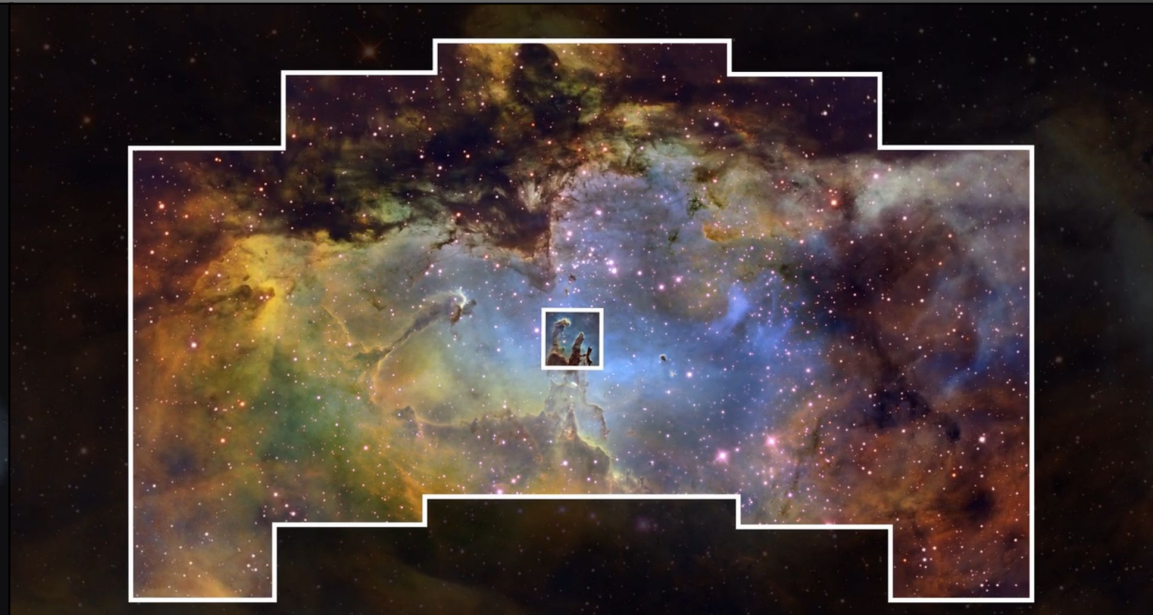
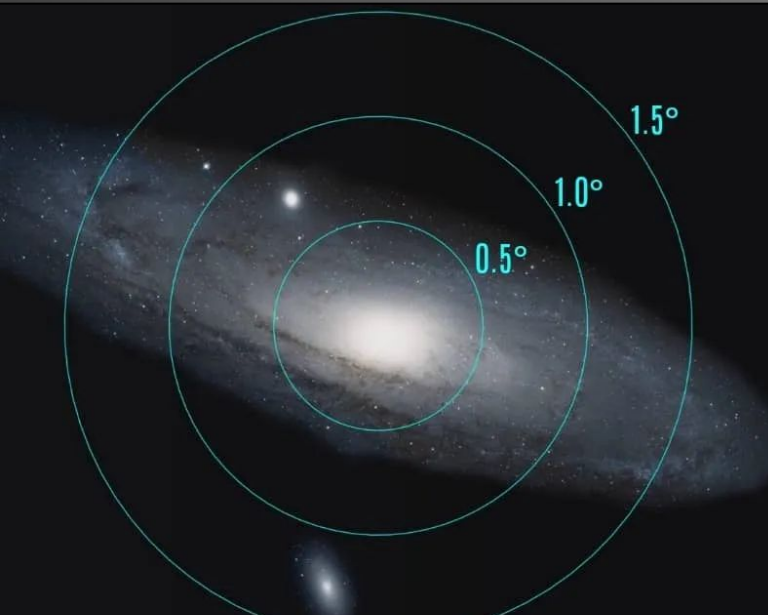
Properties of Telescopes: Focal Length and Magnification

- Focal Length is the distance between the optics, and where the image appears
- We can use the focal length to calculate the telescope's magnification
 - For ASU telescopes:
 - Telescope focal length: 2000 mm
 - Eyepiece focal length: various available, between 7 and 40 mm
 - With a 40mm eyepiece, we get a magnification of 2000/40, or 50 times magnification
 - This means that an object viewed through the telescope will appear 50 times larger than it does with the naked eye
- Note that greater magnification leads to a darker image
 - The same amount of light is getting stretched over a larger area

$$\text{MAGNIFICATION OF TELESCOPE} = \frac{\text{TELESCOPE FOCAL LENGTH}}{\text{EYEPIECE FOCAL LENGTH}}$$

Properties of Telescopes: Field of View (FoV)

- How much of the sky do you see through the telescope
 - Typically measured in angular size
- A longer focal length means a greater field of view



Properties of Telescopes: Resolving Power

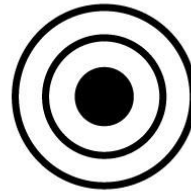
- Also referred to as “Angular Resolution”
- This is the ability to see fine detail or separate two nearby objects
- It is expressed as the minimum (angular) distance between two objects that you are still able to see them as two separate objects
- Applies to things like stars that are close together, or the smallest craters visible on the moon

$$\alpha = \frac{4.56}{D},$$

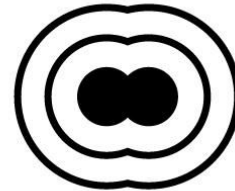
(Dawes limit for a human eye through a telescope eyepiece)

D = diameter of primary mirror in telescope in inches

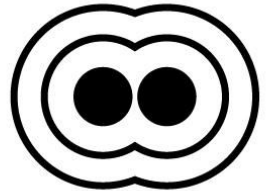
α = angular separation between objects in arcseconds



not resolved



just resolved



well resolved

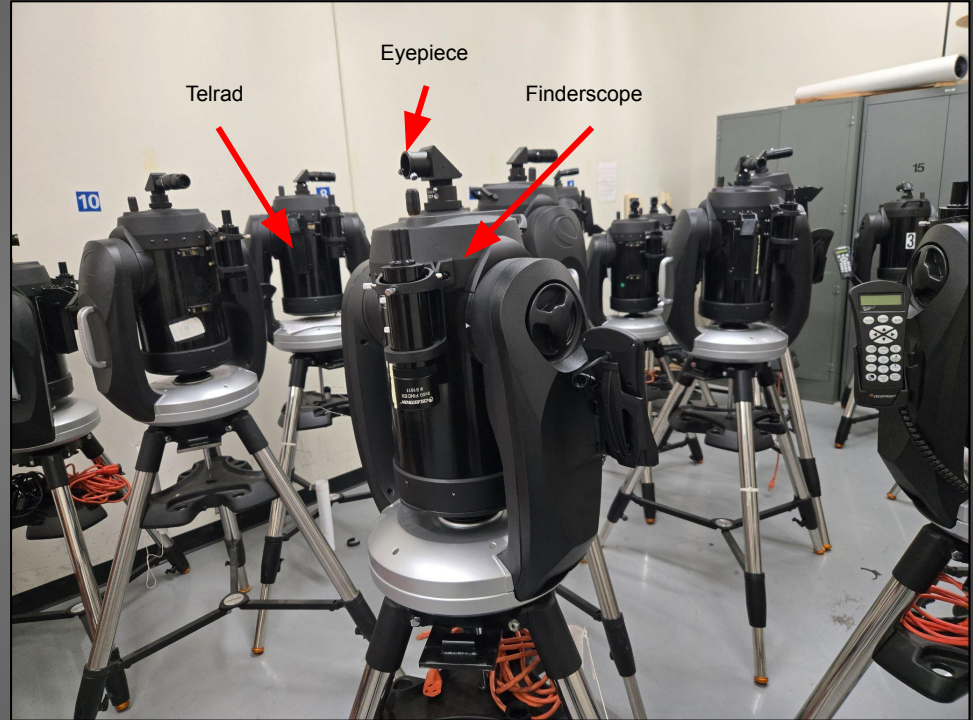
Properties of Telescopes: Scale

- How large an object appears with the telescope compared to without
- It depends on the focal length
- Typically expressed as an angle per length ($^{\circ}/\text{mm}$ or $''/\text{mm}$)
 - Often, with scientific imaging from telescopes we use arcseconds per pixel ($''/\text{pixel}$)

$$\text{scale} = \frac{\text{angular size of an object on the sky}}{\text{size of the final image using the telescope}}$$

ASU's Telescopes

- 8 inch diameter
- 2000 mm focal length



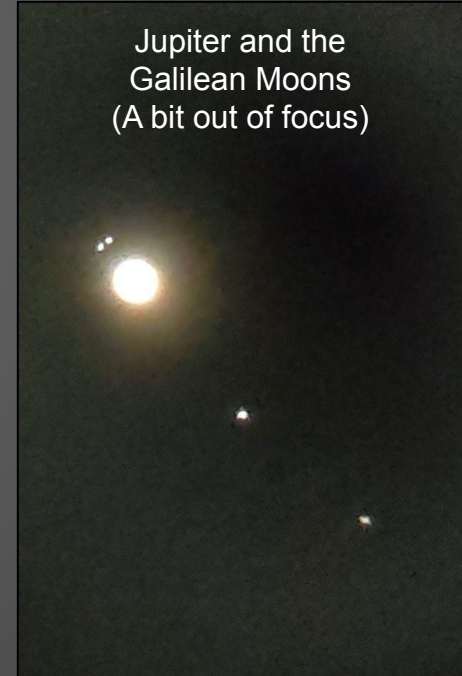
Note: These are not in use, the orientation will be different on the roof.
Additionally, not all of the telescopes have a Telrad.

Tonight's Lab

- Skip question 1, 2, and 5 - you are not aligning the telescopes yourselves
- We will watch a couple videos to complete the table on page 7 and questions 3-4
 - Then we will go to the roof! (if weather is good)



← The Moon



Roof Rules

- Bring your stuff with you
 - We will not be coming back to the classroom
- Do not enter the fenced in area
- Red flashlights only
 - No white lights except in emergencies
- Do not adjust the telescopes!
- We will be up there with one or more other classes
 - You can ask me or the other TA(s) if you have any questions
 - Take turns/work together, no need to look through the telescopes in a specific order (as long as you use the right ones for each question and/or sketch)

Tonight's Lab (On the Roof)

- Do the sketches on page 8 based on what you see in the telescopes
 - Then answer questions 6-11 based on that
- Question 12 is asking about what you think will happen, before you do it
 - Then fill in the table below it and don't miss question 13 and the bottom of the page
- Finish the remaining questions
 - The final answer on question 14 should have units of length (i.e. inches or centimeters)
 - Question 15 is asking about the calculation in question 14, not what we are observing today
 - The "Conclusion" is a series of fill-in-the-blank questions, the options are "increase", "decrease", and "stay the same"

Image Credits (In order of appearance)

- NOIRLab
 - <https://noirlab.edu/public/news/noao9809/>
- Wikipedia
 - <https://en.wikipedia.org/wiki/VERITAS>
- Wikipedia
 - <https://en.wikipedia.org/wiki/Pluto>
- Wikipedia
 - https://en.wikipedia.org/wiki/Large_Binocular_Telescope
- NASA
 - <https://asd.gsfc.nasa.gov/blueshift/index.php/2016/07/25/thirty-years-of-space-vlbi/1000/>
- NASA
 - <https://science.nasa.gov/photojournal/keck-telescope-views-of-uranus/>
- Keck Observatory
 - <https://keckobservatory.org/our-story/telescopes/>
- Astro La Palma
 - <https://lapalmastars.com/en/visit-observatory-lapalma/>
- GTC Gallery
 - <https://www.gtc.iac.es/multimedia/imageGallery.php>

Image Credits (Continued)

- National Geographic
 - <https://www.nationalgeographic.com/science/article/first-images-vera-rubin-observatory-astronomy-space>
- AURA Astronomy
 - <https://www.aura-astronomy.org/centers/nsfs-oir-lab/rubinobservatory/>
- Tenor
 - <https://tenor.com/search/tim-curry-space-gifs>
- Britannica
 - <https://www.britannica.com/topic/Hubble-Space-Telescope>
- National Geographic
 - <https://www.nationalgeographic.com/magazine/article/hubble-telescope-anniversary-10-greatest-images>
- BBC Sky at Night Magazine
 - <https://www.skyatnightmagazine.com/news/james-webb-space-telescope-pismis-24>
- Wall Street Journal
 - <https://www.wsj.com/story/everything-you-need-to-know-about-nasas-james-webb-space-telescope-b00d144e>
- Wikipedia
 - https://en.wikipedia.org/wiki/Nancy_Grace_Roman_Space_Telescope
- NASA
 - <https://science.nasa.gov/asset/webb/hubble-webb-mirror-comparison/>

Image Credits (Continued)

- The lab's own instructions
- Love the Night Sky
 - <https://lovethenightsky.com/what-is-a-field-of-view-in-astronomy-and-why-is-it-useful/>
- ESA
 - https://www.esa.int/ESA_Multimedia/Images/2023/02/Roman_s_field_of_view_compared_to_that_of_Hubble
- The lab's own instructions
- Geeks for Geeks
 - <https://www.geeksforgeeks.org/physics/resolving-power-of-microscopes-and-telescopes/>
- The lab's own instructions
- Taken by me
- Same for the rest